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Article

Cost Analysis of Direct Air Capture and Sequestration Coupled to Low-Carbon Thermal Energy in the United States

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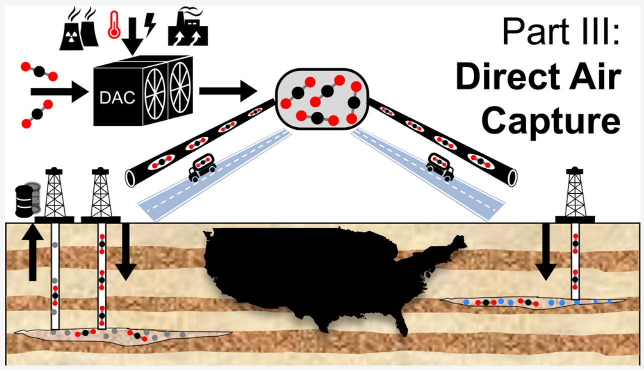
ABSTRACT: Negative emissions technologies will play an important role in preventing 2 °C warming by 2100. The next decade is critical for technological innovation and deployment to meet mid-century carbon removal goals of 10−20 $GtCO_2$/yr. Direct air capture (DAC) is positioned to play a critical role in carbon removal, yet remains under paced in deployment efforts, mainly because of high costs. This study outlines a roadmap for DAC cost reductions through the exploitation of low-temperature heat, recent U.S. policy drivers, and logical, regional end-use opportunities in the United States. Specifically, two scenarios are identified that allow for the production of compressed high-purity CO_2 for costs ≤\$300/tCO_2, net delivered with an opportunity to scale to 19 $MtCO_2$/yr. These scenarios use thermal energy from geothermal and nuclear power plants to produce steam and transport the purified CO_2 via trucks to the nearest opportunity for direct use or subsurface permanent storage. Although some utilization pathways result in the re-emission of CO_2 and cannot be considered true carbon removal, they would provide economic incentive to deploying DAC plants at scale by mid-century. In addition, the federal tax credit 45Q was applied for qualifying facilities (i.e., producing ≥100 $ktCO_2$/yr).



INTRODUCTION

A number of recent studies[1−3] have collectively emphasized the need to include negative emissions technologies in the portfolio of solutions required to prevent 2 °C warming by the year 2100. In particular, the National Academy of Sciences released a report in 2019 that outlined a number of negative emissions technologies that exist today costing less than or equal to \$100/tCO_2 removed and sequestered.[2] Some of these technologies include agricultural soil conservation practices; increased forest management; and waste biomass capture, processing, and distribution. They estimated that to meet global climate goals, 10 billion tonnes of CO_2 ($GtCO_2$) will need to be removed each year up to 2050, with 20 $GtCO_2$ removal each year from 2050 up to 2100. Unprecedented levels of technology adoption and land use changes will be required to meet the short-term goals, whereas costlier or less developed technologies such as direct air capture (DAC), bioenergy with carbon capture and storage, and mineral carbonation will be required to achieve the larger-scale and longer-term removal rates that are anticipated in the second half of the century.[2] However, for the larger scales to be realized by mid-century, deployment on the scale of thousands to millions of tonnes per year needs to take place today.

There are two types of DAC currently being scaled to meet climate goals: (1) chemical liquid solvent DAC (i.e., Carbon Engineering)[4] and (2) chemical solid sorbent DAC (i.e. Climeworks and Global Thermostat).[5,6] Although differences exist, these technologies operate under the same premise: selective removal of CO_2 from ambient air (~400 ppm) by contact with a basic solution (chemical liquid solvents) or a basic modified surface (chemical solid sorbents). The CO_2, now fixated in a carbonate or carbamate bond is liberated from the capture media through application of heat, producing high-purity CO_2 and renewed solvent, or sorbent, capable of many cycles of CO_2 removal. Regardless of technology, the average energy requirement is approximately 80% thermal and 20% electrical.[2]

In the case of chemical liquid solvent DAC, CO_2 is ultimately captured via a carbonate pellet, which requires high-

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quality heat at 900 °C[7] for regeneration of the capture material and subsequent production of high-purity CO_2. Unfortunately, heat of this quality is not available in industrial or power generation waste streams. Alternatively, solid sorbent regeneration requires much lower quality heat at approximately 100 °C, which may be accessible as waste heat in a variety of processes. This indicates that coupling solid sorbent DAC with waste heat sources has the potential to offset 80% of the process energy requirements. For these reasons, this study targets energy partnerships. Specifically, the use of waste heat from geothermal energy or a slip-stream from nuclear energy production to assess the viability of these low-carbon and low-cost heat sources for modular solid sorbent-based DAC. This can provide a necessary piece of the pathway required for immediate, cost-effective deployment of solid sorbent DAC.

The scaling of DAC must start now and incentives to scale must be developed in tandem with deployment. To date, Climeworks has the only commercial-scale solid sorbent DAC plants worldwide collectively removing several thousands of tonnes of CO_2 with demonstrated costs at roughly \$600/tCO_2.[5,8,9] To achieve the required scale outlined in the National Academies report, DAC plants must be deployed in the near-term as the deployment of more plants could lead to learning by doing and subsequent cost reductions.[2]

To achieve the removal targets of $GtCO_2$ per year by midcentury would require an annual growth of 45% that is maintained over the next 30 years. Within the framework of energy-technology deployment, scale-up typically takes place at a rate of 1 order of magnitude per decade, which corresponds to a 26% annual growth rate.[10] For illustration, the growth and experience rates of industries such as photovoltaics (PV) can be examined as a model for achieving the type of cost reductions and deployment scaling required for DAC to make meaningful contributions to climate change (see the Supporting Information, Experience Curves). Using the accelerated growth rates seen with PV (70% from 2000 to 2010 and 60% from 2010 to 2016),[11,12] DAC would achieve a scale on the order of 80 $MtCO_2$/year by 2050 with the ability to achieve $GtCO_2$/yr scale by 2060.

These growth models show optimistic potentials reaching 10s of $MtCO_2$/year by mid-century and policy will be crucial to making larger facilities and standardized manufacturing financially viable. Although niche markets (such as aerospace and consumer products) allowed for the initial deployment of PV to 40 MW,[12] policy incentives are one key reason why PV was able to scale so quickly. For example, a study carried out by Kavlak et al.[13] indicated that government policies such as renewable portfolio standards, feed-in tariffs, and a variety of subsidies accounted for roughly 60% of the market growth of PV. These subsidies further allowed for continued and rapid technology innovation. As a given technology scales, there may be improvements that require redesign and system re-optimization. Currently, DAC is at the scale of thousands of tonnes of CO_2 per year and further deployment at this scale will outline the research and development (R&D) needs required so that advanced and more cost-effective processes may be adopted.

Recent federal policy developments have provided new financial drivers for scaling and optimizing DAC projects. The reformed 45Q tax credit within the 2018 Bipartisan Budget Act[14] provides a tax credit of \$15.29/tCO_2 for enhanced oil recovery (EOR) and utilization in 2018, increases linearly to \$35t/CO_2 in 2026, and rises with inflation thereafter. The Act limits qualified projects to those that have commenced construction prior to 2024, which could be extended in future budgets. Furthermore, DAC facilities must capture at least 100 $ktCO_2$/yr and the credits can only be claimed for 12 years; however, there is no limit on the number of total or project bankable credits. This policy is designed to provide substantial interim incentives to increase both the efficiency and scale of DAC projects. Currently, the uncertainty of the extension of the 45Q tax credit for projects beyond 2024 makes the development of future projects economically ambiguous. Nonetheless, the transferability of the 45Q credit makes it possible for DAC companies and EOR or utilization operators to distribute credits and develop early partnerships, while scaling and improving their technologies.

Until mass carbon markets can develop, DAC must be sustained by policy efforts as well as smaller, niche markets and customers. However, the best opportunities for DAC to begin commercializing today do not necessarily align with the ultimate markets for DAC as a negative emissions technology, that is, permanent storage. For one, it is important to find a robust paying customer, whether through existing markets for CO_2 utilization and/or policy incentives for DAC. In many cases, these initial markets will not involve negative emissions, yet still provide value in facilitating the goal of technology cost reductions through initial deployments. However, through robust policy incentives, economic pathways for CO_2 storage can be developed to achieve negative emissions.

Although utilization or recycling of CO_2 may seem like a distraction from the primary goal of the permanent removal of CO_2 from the atmosphere,[15] there are scenarios in which utilization could serve as a bridge toward DAC becoming a negative emission technology. For instance, lowering barriers of investment into the construction of DAC plants on pilot- and demonstration-scales would lead not only to advanced designs but also to an increased demand in the sorbent material and equipment required. An increase in demand could stimulate competition in the field and advance R&D efforts, ultimately yielding reduced systems costs. The learning one acquires by building and operating DAC facilities under realistic conditions will ultimately provide greater clarity of the investments required and the specific areas of focus that will require further R&D.

The current study investigates pathways toward jumpstarting the deployment of DAC. First, energy partnerships between geothermal and nuclear energy producers and solid sorbent DAC are suggested. Here, the cost of sorbent DAC is determined, including access to waste heat from geothermal or a nuclear slip stream. The cost of truck transport for the removed CO_2 is also included to give a cost of CO_2 net delivered to a site for end use (geological storage or utilization), accounting for all CO_2 emissions involved in the process. From here, the focus is shifted to locational synergies and optimal locations for first-of-a-kind solid sorbent DAC plants. By focusing on identifying markets and opportunities that have the greatest ability to reduce DAC technology costs in the near-term (such as EOR), the DAC industry can begin learning by doing and work toward the necessary scale of $GtCO_2$ per year by mid-century. Specifically, this analysis focuses on cost reduction using existing power generation infrastructure. Coupling DAC to a low-carbon and low-cost thermal source while considering the most proximal utilization or geological sequestration site allows for the realization of the lowest-cost pathways.

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■ METHODS

The analysis presented here focuses on solid sorbent DAC systems. These systems are highly complex and are composed of many parameters that must work together in an integrated fashion. Therefore, many parameters can vary between similar technologies. For example, the structure of the contactor (in this case, a monolith) will directly influence the pressure drop across the system, which in turn influences the required fan power and electricity requirements. Other such factors include the sorbent CO_2 capacity, the desorption swing capacity, and the adsorption and desorption cycle times, among others. As the sorbent-based DAC is a relatively new technology, a general configuration is built for this analysis from the available literature.[2,16−18] Common to all solid sorbent DAC systems is a contactor composed of functionalized sorbent and monolith, fans, and vacuum and compression pumps.

The solid sorbent system analyzed here is based on a 5-step temperature vacuum swing adsorption process, as described in the National Academies of Sciences 2019 study.[2] Specific parameters and detailed equations used to build the sorbent DAC model are presented in the Supporting Information. It is assumed that 400 ppm CO_2 under ambient conditions enters each 1.1 m × 1.1 m contactor unit at 3.0 m/s, where fan power operating at 0.70 efficiency helps to overcome the 480 Pa pressure drop. For ambient conditions, 3.0 m/s is considered optimal across the sorbent bed, as higher velocities do not show a significant impact on adsorption rate.[18] Furthermore, the adsorption cycle is assumed to last 20 min and CO_2 is selectively adsorbed at a rate of 1.0 mol/kg sorbent. During the desorption process, the temperature is at 373 K and the overall desorption swing capacity is 0.8 mol desorbed/mol adsorbed.

Siting DAC Plants Based upon Available Low-Carbon and Low-Cost Thermal Energy. Here, two routes are explored in which low-cost, low-carbon heat satisfies the 80% sorbent DAC thermal requirement: (1) heat extraction from the exit brine of existing geothermal power plants, ideally placed post-power generation and pre-reinjection and (2) a 5% slipstream of steam from a nuclear power plant steam generator is diverted to the DAC plant and returned to the nuclear plant upstream of the moisture separator−reheater. These scenarios are compared against a base case in which steam produced from natural gas is used to meet the thermal energy requirements. In each of the two scenarios, the steam or warm water flow rate from each facility was used to determine the size of the DAC plant in terms of CO_2 removal per year. The thermal energy assumed was based upon a generic adsorption process representing 80% of the total energy at 6 GJ/tCO_2. The electrical energy associated with running fans and pumps was assumed to comprise 20% of the total energy at approximately 1.5 GJ/tCO_2.[2]

Compared to the base case, it was assumed that additional capital was required to integrate DAC with geothermal and nuclear heat. Because of the corrosive nature of geothermal fluids and to account for possible scaling of silicates at lower temperatures, the capital expense for the heat exchanger (the only DAC component in direct contact with the geothermal fluid) is tripled to account for equipment replacement over the lifetime of the project. In addition, plant modification costs are built into the cost estimates for both nuclear and geothermal configurations (Table S1). The infrastructure required to produce steam for solid sorbent regeneration was not included because the particular units considered are electric utilities and therefore already have the infrastructure in place. Details about heat recovery and process flow sheets are provided in the Supporting Information for each scenario.

Base Case: Use of Steam Derived from Natural Gas. The base case is a system in which thermal energy for DAC was supplied via steam from natural gas without point-source capture. Therefore, the emissions from natural gas combustion are included in the analysis.[19,20]

Scenario #1: Use of Waste Heat from Geothermal-Based Electric Utilities. The operating capacity of geothermal power reached approximately 13 GW globally in 2015, with 28% taking place in the United States.[21] On average, geothermal electric stations produce approximately 45 gCO_2-equiv per kW h generated,[22−24] which is less than 5% of that produced by conventional coal-fired power plants. The colocation of DAC plants with geothermal would add further benefit as a low-greenhouse gas emission approach for power generation. Current state-of-the-art conversion technologies include dry steam, flash steam, and binary cycle, spanning over a broad temperature range of the geothermal working fluid. In this study, only currently operating geothermal power plants are considered for siting the DAC plants. Information on geothermal power plants in the United States has been acquired from the NREL geological prospector,[25] including the location, conversion type, and power capacity. The steam or warm water flow rate was estimated based upon the type of power plant and its capacity.

Scenario #2: Use of 5% Slip Stream of Steam from Nuclear-Based Power Plants. The operating capacity of the nuclear power fleet reached approximately 400 GW_e globally in 2018, with 25% taking place in the United States.[26] Because of the sub-threshold temperature of the plant exhaust (e.g., <50 °C), the DAC plant is integrated to utilize 5% of the steam immediately following the steam generator at 6.0 MPa and 275 °C through direct piping to the adjacent DAC facility. This steam is returned via piping to the nuclear plant, where it rejoins steam exiting the high-pressure turbine and is reheated and passed downstream to the array of low-pressure turbines. Assuming the high-pressure turbine system is responsible for 35% of the plant generating capacity, diversion of 5% of steam at this step will compromise the electricity generation in the nuclear plant by roughly 1.5%.[27] This loss in potential revenue represents the steam operating expense. Information regarding operating nuclear power plants was retrieved from the U.S. Nuclear Regulatory Commission,[28] as well as public design documents from Westinghouse.[29]

Developing the Net Delivered Cost. In the current work, the cost of DAC versus the "net delivered" cost of DAC are distinguished. Here, the net delivered cost is the cost associated with delivering one tonne of compressed CO_2 sourced from the atmosphere to its respective end use or storage location. In calculating the net-delivered cost, all related CO_2 emissions must be accounted for from the system. This includes embodied emissions in materials, indirect emissions associated with power and heat requirements, as well as those associated with compression and transport. For this analysis, a cradle-to-gate lifecycle model is used. Although emissions associated with the DAC process, compression, and transportation are included, the downstream emissions from either the subsequent utilization of CO_2 or the CO_2 storage process are not included in the analysis. For the base case, the emissions associated with upstream natural gas leakage are also neglected.

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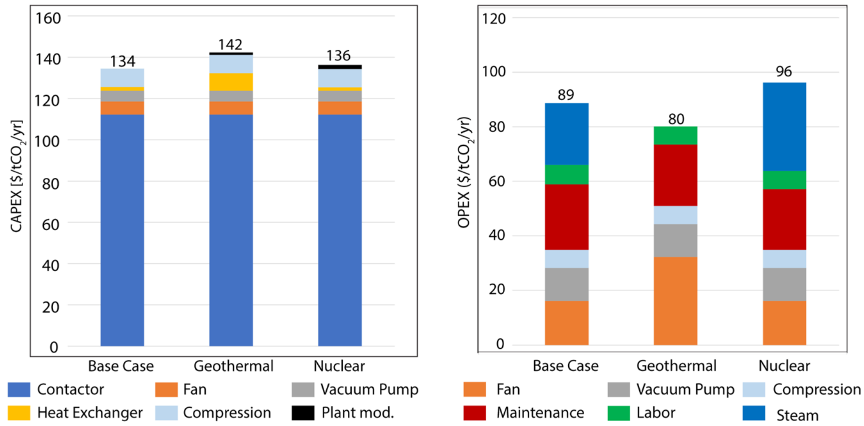

Figure 1. Breakdown of annualized capital cost (left) and annual operating and maintenance costs (right) per tonne of CO_2. Costs are calculated for a system capable of processing 100 ktCO_2 per year.

The cost of DAC was developed using the sorbent configuration described previously. Additional cost considerations for the delivered costs include compression of the produced CO_2 and transportation of the CO_2 to a site for geological storage or utilization. The compression model is based on the work of McCollom and Ogden.[30] A 5-stage compressor train with dehydration and inter-stage cooling is employed to prepare the CO_2 stream for trucking transport (1.7 MPa and −30 °C). The inlet temperature at each stage is 343 K with compression ratio of 1.76 and isentropic efficiency of 0.75. Cooling to final transport temperature is achieved at 16.5 kW h/tCO_2. The transport model is based heavily on the work of Berwick and Farooq[31] and takes as inputs the overall volume of CO_2 to be delivered per route and total distance per route. Further details can be found in the Supporting Information.

The net delivered cost includes all of the direct and indirect emissions for the DAC system, compression, and transportation. Furthermore, as the removal of CO_2 from air requires significant contactor surface area and materials, embodied emissions were also included in the estimate. However, it was found that the embodied emissions represented less than 1% of the total cost of removal. When low-carbon energy resources are used for DAC and all steps of the process are colocated, the captured cost and net delivered cost begin to converge. Considering that DAC plant energy is dominated by thermal contributions, identifying low-carbon heat sources is critical to minimizing net delivered costs.

For the net delivered cost to be transformed into the net removed cost, the CO_2 must be permanently stored, that is, through sequestration in a viable underground storage site or through the reaction with suitable inputs to form stable carbonates, and the CO_2 emissions from this storage process must be accounted for. Accordingly, utilizing captured CO_2 for applications such as fuel synthesis or for beverage carbonation does not yield negative emissions. Primary methods of permanent sequestration considered in the current study include storage in geological formations, either through injection in saline aquifers or in depleted oil and gas reservoirs.

The economics in this study are based on a 12.5% discount rate and 10-year DAC economic lifetime. The discount rate was back calculated from the Integrated Environmental Control Model,[32] where it was assumed that the risk profile of sorbent DAC is similar to point source carbon capture technologies. The economic lifetime is used to coincide with the monolith lifetime,[18] but also affords DAC facilities deployed early flexibility to re-evaluate siting as conditions dictate. Here, the discount rate and plant lifetime results in a capital recovery factor of 18.39%. This is slightly higher than other analyses evaluating the cost of DAC, including those using capital recovery factors near 7.5−12.5%.[2,7,33]

Given the modular nature and mobility of the sorbent-based DAC process, there is less risk in projecting future scenarios whereby either the heat source becomes unavailable (e.g., fleet expiration) or better opportunities exist elsewhere (e.g., paying customers surface, government procurement). In fact, some current designs use shipping containers to house the contactor, making relocation of a DAC plant based on this technology easier than in cases of fixed infrastructure.

Geological Sequestration. Data regarding CO_2 storage were provided by the USGS Assessment of Geologic Carbon Storage and was used to estimate the storage capacity of each reservoir as well as to map the proximity of the storage sites to the geothermal and nuclear facilities.[34,35] Further details of this analysis can be found in the Supporting Information.

RESULTS AND DISCUSSION

Cost Estimates of Each Scenario at 100 ktCO_2 per Year. For solid sorbent DAC, the dominant capital component is the contactor, primarily because of the need to replace the deactivated sorbent yearly over the lifetime of the plant. As indicated in Figure 1, this single item constitutes roughly 80% of the annualized capital cost where the sorbent lifetime is assumed to be 1 year with a cost of $50/kg (Table S1). Compression does not change as each system described in Figure 1 delivers the same volume of purified CO_2 product to the compression unit.

Operational and maintenance costs are generally dominated by fan power, which is used to overcome the system pressure drop.[36] As the contactor is consistent among all of the systems examined, the pressure drop does not change. A similar relationship is observed in the vacuum operating costs, and it should be noted that the channel gas velocity should be optimized so that the entire sorbent bed is exposed to CO_2. Finally, higher operational costs observed for the base and nuclear cases are tied to steam OPEX, which are derived from the energy penalty to the steam provider as $2.8 and $3.9/GJ for the base and nuclear cases, respectively.

This cost analysis is presented in terms of DAC capacity or the amount of CO_2 captured per year. Within this metric, the

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total cost of capture (capital and operating) is \$223, \$205, and \$233/tCO_2 captured for the base, geothermal, and nuclear cases, respectively. Given the similarity in capture costs for the base case, geothermal, and nuclear systems, there are no apparent cost savings to offset the apparent risks in engineering these systems.

However, because of the incorporation of low-carbon heat, the DAC emissions footprint for these configurations is reduced from 0.65 tCO_2 emitted for every tCO_2 captured for the base case to 0.29 tCO_2 emitted for every tCO_2 captured for both geothermal and nuclear configurations. The importance of considering both captured cost and process emissions footprint when discussing DAC is evident in Figure 2. The net delivered cost is highly dependent on the process carbon intensity (CI). For example, with a base case footprint of 0.65 tCO_2 emitted for every tCO_2 captured, only 0.35 tCO_2 is removed and the cost to remove and deliver 1 tonne is almost 3× the reported capture cost of \$223/tCO_2. These footprints are calculated based on a heat intensity of 227 gCO_2/kW h from natural gas[2] and a grid intensity of 490 gCO_2/kW h (based on the current US mix of electricity generation by source: 35% natural gas, 27% coal, 19% nuclear, and 17% renewable[37]). Further reductions in the net delivered cost can be achieved through use of low-carbon electricity.

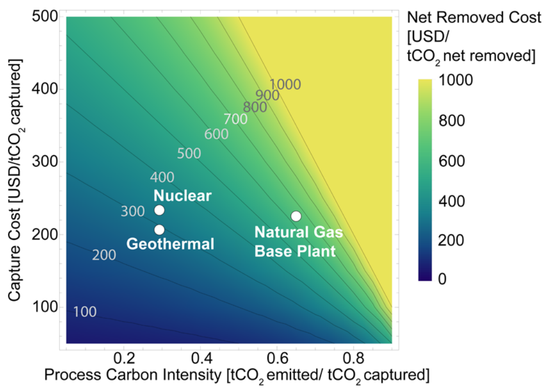

Figure 2. Impact of captured cost and process CI on the net delivered cost of CO_2. Although all configurations considered have similar capture costs, the net delivered cost is heavily reliant on the process carbon footprint, calculated here from a grid intensity of 490 gCO_2/kW h and heat intensity of 227 gCO_2/kW h (from natural gas). The incorporation of low carbon heat results in savings on the order of hundreds of dollars over the base case. Costs are calculated for a system capable of processing 100 $ktCO_2$ per year.

Scaling the Costs of DAC. The cost analyses in this study are presented for configurations that capture 100 $ktCO_2$ per year. However, many early-entry systems are expected to be much smaller; thus, it is important to understand how these costs scale. The solid sorbent contactor employs a modular design which lends to linear scaling of costs, operational flexibility, and the opportunity to standardize process manufacturing. Owing to the modular contactor design and dominating capital share, the annualized capital costs remain relatively flat across system size. Although other process equipment such as fans, pumps, and turbines follow scaling laws with equipment-specific scaling factors, these economies of scale are largely masked due to the smaller contribution of these components to the overall capital. This can be seen in Figure 1 given the small contribution of the fans, pumps, plant modification, and similar equipment to the overall system capital costs.

Extension of this cost breakdown to total capital required warrants a discussion of general plant scaling rules. The common 2/3 law states that the cost of plant with power output, P_i is related to a reference plant with power output, P_o and cost, C_o as shown in eq 1

$$C_i = C_o\left(\frac{P_i}{P_o}\right)^{2/3} \tag{1}$$

This relationship adequately captures the nonlinear scaling relationship of a large majority of processing equipment.[38,39] However, in a modular DAC plant where the dominant cost component is expected to obey more linear scaling laws, the power factor should approach unity. Using this approach and assuming a 10 $ktCO_2$/yr plant for reference, the conservative total capital requirement is roughly \$8.5M for both geothermal and nuclear configurations.

Cost Subsidies through Geological Storage and Utilization. DAC facilities that remove 100 $ktCO_2$/yr or more from air qualify for tax credits under the federal tax code 45Q. There are several caveats worth noting. First, these credits only apply to facilities that commence construction by 2024. Second, the captured CO_2 must be paired with geological storage or beneficial reuse, which is defined to cover chemical fixation and conversion of CO_2 as well as use in any existing commercial markets. However, it is unclear how the lifecycle emissions of certain reuse opportunities apply to the net qualifiable CO_2. For these reasons, this analysis is restricted to 45Q tax credits applied to geological storage and EOR. Beverage carbonation is included as a third option outside of 45Q. Although the beverage industry is roughly 30× smaller than CO_2-EOR, the cost of CO_2 for this application is significantly higher. A discussion of revenue possibilities for CO_2 resale in the food and beverage markets is presented in the Supporting Information.

Basin-level geological storage opportunities, as well as CO_2-EOR opportunities are mapped in Figure 3. Under the constraints of 45Q, roughly 16 $MtCO_2$/yr qualify for the tax credit with geothermal and nuclear DAC facilities making up 65 and 35% of the total, respectively. If the plant threshold were lifted, exploitation of all DAC opportunities with geothermal waste heat and a 5% nuclear slip stream would result in the atmospheric removal of 19 $MtCO_2$/yr. This includes an additional 2.6 $MtCO_2$/yr from 61 geothermal DAC facilities and 0.4 $MtCO_2$/yr from 5 nuclear DAC facilities. Decreasing the 45Q threshold to 50 $ktCO_2$/yr would qualify roughly 72% of these additional opportunities for the tax credit, resulting in an increase in removal of 2.2 $MtCO_2$/yr. Here, 81% of the increase is from 25 geothermal DAC facilities and 19% is from 5 nuclear DAC facilities.

Realization of DAC with Geological Sequestration as a Negative Emissions Technology. Ideal storage sites offer high levels of permeability for rapid and high-volume injection and cap rock for the prevention of leakage. Interaction mechanisms in the reservoir between the rock and the CO_2 may help for long-term trapping of CO_2 in the reservoir. Trapping mechanisms include solubility trapping (dissolution of CO_2 into the brine), residual gas trapping (immobilization by capillary forces in the postinjection period), and mineralization through geochemical interactions between the CO_2, brine, and rock.[2] The USGS assesses the possibility of CO_2 sequestration

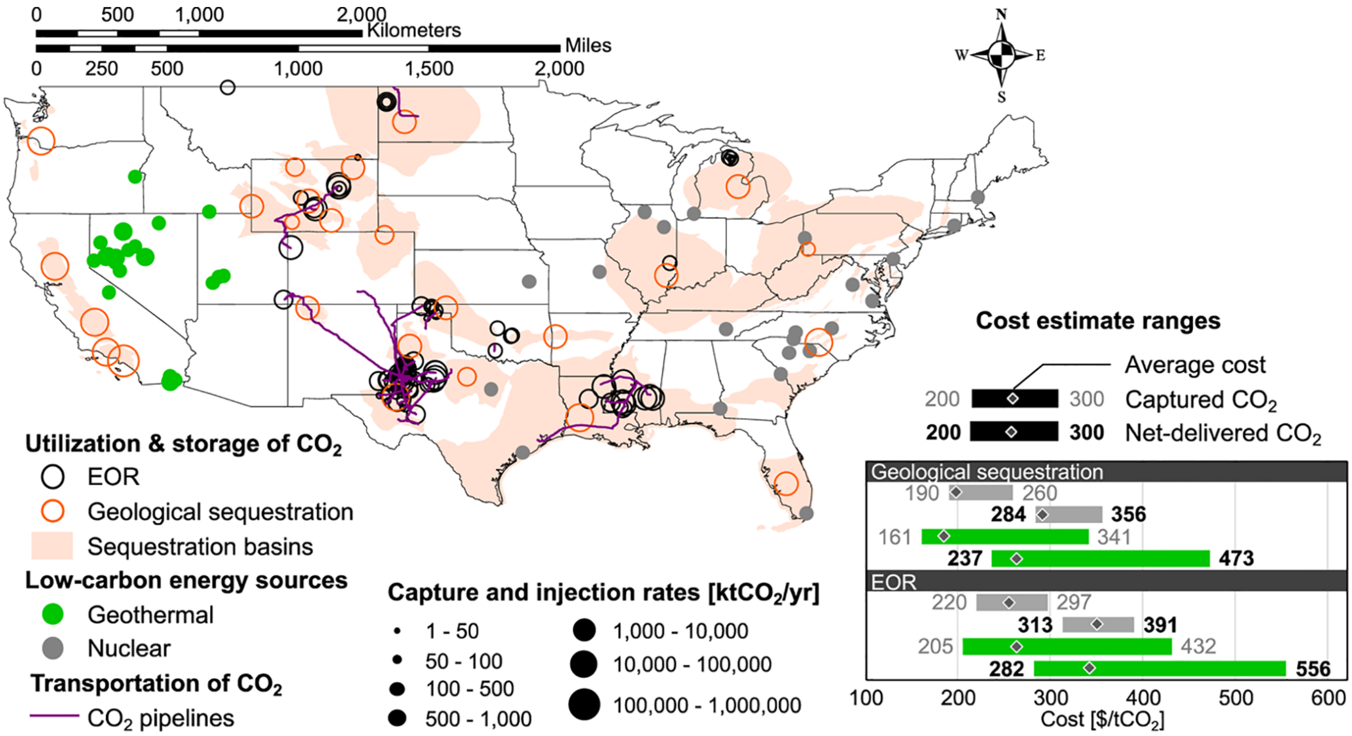

Figure 3. Geographical distribution of the proposed scenarios along with the estimated ranges of the costs. The map shows the distributed geological sequestration (open orange circles) and EOR (open black circles) opportunities with DAC plant possibilities based upon the scenarios of using available thermal energy sourced from geothermal (closed green circles) and nuclear (closed gray circles) for opportunities qualifying for 45Q tax credit (100 ktCO_2/yr or greater). The cost figure shows the projected costs per tonne of CO_2 delivered for DAC paired with reliable storage and reuse (EOR). Included is the capital, operating, and maintenance expenditures for capture, separation, compression, and transport for all scenarios, plus injection in the nearest storage basin for the geological sequestration option and less tax credits for qualifying facilities as described under 45Q.

in contiguous U.S. basins, based on the characteristics of the reservoir and the existing cap rock. Capacities range from 230 MtCO_2 for the Eastern Great Basin under Nevada and Utah to 1800 GtCO_2 for the U.S. Gulf Coast Basin under Alabama, Arkansas, Louisiana, Mississippi, and Texas (Figures 3 and S6). Cumulative storage potential is estimated at 2740 GtCO_2 in the contiguous United States.[1]

The contiguous United States has 96 geothermal and 41 nuclear power plants with the potential to capture 12.8 and 6.0 MtCO_2/yr, respectively. Of these facilities, 35 geothermal and 36 nuclear power plants qualify for the 45Q federal tax credit, accounting for 10.2 and 5.6 MtCO_2/yr. No geothermal power plants are located above sedimentary basins suitable for CO_2 sequestration due to the incompatibility between good geothermal resources and sedimentary basins suitable for CO_2 sequestration. Here, 37 geothermal plants with a capture potential of 3.4 MtCO_2/yr are less than 100 miles from a sedimentary basin, of which 9 plants (with a capture potential of 1.9 MtCO_2/yr) are eligible for 45Q. An additional 49 geothermal plants with a capture potential of 7.1 MtCO_2/yr are between 100 and 200 miles from a sedimentary basin, of which 20 plants (with a capture potential of 6.1 MtCO_2/yr) are eligible for the 45Q tax credit. Most of nuclear power plants are located above or within 100 miles of sedimentary basins suitable for CO_2 injection. Here, 15 nuclear DAC plants, all eligible for 45Q, with a capture potential of 2.3 MtCO_2/yr are colocated to sedimentary basins suitable for CO_2 injection. Of the remaining nuclear plants, 23 plants with a capture potential of 3.3 MtCO_2/yr are less than 100 miles from a sedimentary basin, of which 20 plants (with a capture potential of 3.1 MtCO_2/yr) qualify for the 45Q tax credit.

Qualifying DAC facilities that participate in CO_2 storage would receive a tax credit rate up to \$50 per tonne under 45Q. Figure 3 shows the total range of costs associated with CO_2 removal, compression, and transport to the most proximal geological storage site as depicted in the map in the same figure. This cost also includes the cost of injection into the reservoir. The three costs presented in Figure 3 include an average cost, the net delivered, and the cost of capture. The cost of capture and net delivered costs are discussed in more detail in the Methods section.

The lower end of the nuclear scenario is a cost of roughly \$284/tCO_2 net delivered, whereas the lower end of the geothermal costs is roughly \$237/tCO_2 net delivered. The bulk of this cost discrepancy is primarily attributable to the additional steam OPEX for nuclear-based DAC plants, which adds approximately \$55 per t/CO_2 net delivered relative to a geothermal plant (using a 100 ktCO_2/yr plant size basis). Additional discrepancies can be attributed to site proximity relative to reliable storage sites and, to a lesser extent, system size. For example, the lowest cost option for a nuclear facility incurs a transport cost of \$12/tCO_2, compared to \$18/tCO_2 for the best-case geothermal option. Here, locations of the basins are assumed to be the centroid for calculating the trucking distances and the cost of CO_2 net delivered. However, the edges of the basins are on average 80 miles from the centroid and up to 500 miles for the largest basin (U.S. Gulf Coast basin). As described previously, the modular nature and dominant capital share of the contactors masks economies of

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scale for larger systems. However, system size becomes increasingly important as plants move into eligibility for tax credits via policy mechanisms such as 45Q.

Using Atmospheric CO_2 in Commercial Markets. Approximately 70 $MtCO_2/yr$ (~1.5% of U.S. emissions in 2017) is produced either naturally (55 $MtCO_2/yr$) or anthropogenically (14 $MtCO_2/yr$) to meet U.S. utilization needs. Outside of EOR (85% of total demand),[40−42] the uses are smaller in scale and are generally categorized as chemical, for example, as feedstock for fertilizer (4.3 $MtCO_2/yr$)[43,44] or methanol (2.5 $MtCO_2/yr$),[45−47] or physical use, for example, as a solvent, refrigerant, or in carbonating beverages (3 $MtCO_2/yr$).[48]

Current U.S. EOR practices would not be considered negative or neutral because roughly 85% of the CO_2 used for EOR is sourced from natural CO_2 reservoirs.[49] Alternatively, if the CO_2 is sourced from DAC and if there was economic incentive to maximize the CO_2 left in the earth, CO_2-EOR could serve as a building block to an integrated system designed to achieve negative emissions.[40,50,51] An example of this type of integrated system is advanced EOR, which serves to maximize CO_2 storage. An analysis by Núñez-López et al.[52] evaluated four different EOR methods [continuous gas injection (CGI), water curtain injection (WCI), water-alternating gas (WAG), and a hybrid approach between WAG and WCI]. Here, it was determined that the first several years of operation can yield "negative" emissions (13−16 years for CGI, 6−6.7 years for WCI, 14−18 years for WAG, and 6−6.4 years for the hybrid). These values correspond to 24−72% of the project lifetime, depending on the process.[52] In a related analysis, Núñez-López et al.[52] evaluated these same alternatives in a scenario similar to maximum storage EOR. Here, it was determined that EOR operation can yield negative emissions for 22.5 years to project lifetime for CGI, 6−9 years for WCI, 19 years to project lifetime for WAG, and 9−12 years for the hybrid scenario.[52] It is important to note however, that the carbon atom density in condensed CO_2 at the temperature and pressure conditions of the earth will always be lower than the carbon atom density of oil. Hence, to truly put more carbon back in the earth than is produced through the recovery and burning of oil requires not just the displacement of oil for CO_2, but also dedicated storage projects where additional CO_2 is permanently sequestered. Further analysis is required to determine the true negative emissions potential of joint DAC−EOR systems. This is important not only for climate impact but also to maximize 45Q benefits. To be considered competitive, DAC has to compete with incumbent CO_2 providers to EOR fields, which currently stand between \$10−40/tCO_2[40,51,53] and are largely sourced from natural CO_2 reservoirs (e.g., McElmo Dome and Jackson Dome).

Taking into account all DAC facilities that qualify for 45Q, those paired with beneficial CO_2 reuse opportunities would earn a tax credit rate of up to \$35 per tonne. Figure 3 shows the range of costs associated with CO_2-EOR opportunities. The average nearest EOR opportunity is 468 and 680 miles for nuclear and geothermal facilities, respectively.

Considering the cost of capture and transport for existing modular DAC systems, additional policy incentives at the state level will likely be necessary to optimize and scale future DAC facilities.[54] California's Low Carbon Fuel Standard (LCFS) places a cap on the maximum CI of transportation fuels sold within the state and grants credits for fuels below the CI requirement. Recently, the LCFS has also incorporated a protocol to provide credits to "an entity that employs DAC to remove CO_2 from the atmosphere and geologically sequester the CO_2."[55] The protocol defines geological storage as storage within saline aquifers or depleted oil and gas reservoirs, including for EOR and EGR, anywhere in the world. In the year prior to July 2019, the credit traded at an average between \$172/tCO_2 and \$188/tCO_2. In combination with the 45Q tax credit, the LCFS could provide a powerful incentive for EOR projects to partner with DAC companies.

Paths toward Accelerating the Deployment of DAC. Identifying low-cost DAC opportunities as outlined in the current work will enable the formation of partnerships that could lead to deployment on the 100s to 1000s of $ktCO_2/yr$ removal scale. Coupling DAC with available low-cost and low-carbon thermal energy sources will assist in initial costs reductions. Combined with policy incentives, these pathways can lead to a deployment of DAC that has the greatest climate impact. The partnership opportunities outlined in the current work have the opportunity to remove up to 19 $MtCO_2/yr$ in the United States for as low as \$200−\$300/tCO_2 net delivered. Recent partnerships between Coca-Cola Company subsidiary, Valser, and Climeworks,[56] Global Thermostat and ExxonMobil,[57] and Carbon Engineering and Oxy Low Carbon Ventures,[58] a subsidiary of Occidental Petroleum, demonstrate increasing collaboration between the beverage and petroleum industries and DAC companies. The value of policy incentives, such as the 45Q tax credit and California's LCFS, in creating a new carbon market is continuously demonstrated by new and developing partnerships formulated to capitalize on these incentives. Other recent collaborations include Climeworks' partnerships with Audi[59] and Rotterdam The Hague Airport[60] to synthesize fuels from atmospheric CO_2.

Through strategic partnerships between DAC companies and more mature utilization and EOR markets, DAC manufacturers can improve and scale their technologies, while also benefiting from policy incentives and cost-reducing technology couplings. As these low-cost pathways are realized, it is increasingly necessary to look ahead to ensure progress and avoid the situation whereby niche markets become saturated and demand stalls. Future work will consider additional geothermal opportunities in addition to industrial waste heat opportunities, which may collectively lead to $GtCO_2$ removal potential, provided they are fully exploited. Overall, we must utilize these low-cost, low-carbon pathways to deploy DAC today, while we develop policy incentives that drive scaling and push DAC down the learning curve and encourage continued partnerships.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.0c00476.

Cost model: baseline assumptions; details of experience curve model; experience curves; plant configurations; equipment scaling and DAC capacity; operations and maintenance; calculating DAC capacity; cost of net delivered CO_2; government incentives include the federal tax credit 45Q and California's LCFS and both may be applicable to DAC; geothermal power plant scenario details; nuclear power plant scenario details; additional assumptions and boundary conditions associated with the use of solid sorbents for DAC; utilization opportunities at incumbent price points; although small-

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scale, the bottled CO_2 market could offer cost offsets; learning through CO_2-EOR; geological sequestration; and trucking transportation is the most cost-effective for scales <500 $ktCO_2$/yr (PDF)

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Notes

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ABBREVIATIONS

BECCS	bioenergy with carbon capture and storage
CAPEX	capital expense
CGI	continuous gas injection
CHP	combined heat and power
CI	carbon intensity
CO_2-EOR	EOR using CO_2
DAC	direct air capture
DACS	DAC with geological sequestration
EGR	enhanced gas recovery
EOR	enhanced oil recovery
$GtCO_2$	billion metric tonnes of CO_2
ISBT	international society of beverage technologists
$ktCO_2$	thousand metric tonnes of CO_2
LCFS	low carbon fuel standard
$MtCO_2$	million metric tonnes of CO_2
NREL	national renewable energy laboratory
OPEX	operating expense
PV	photovoltaic
PWR	pressurized water reactors
SAU	storage assessment units
tCO_2	metric tonnes of CO_2
TVSA	temperature-vacuum swing adsorption
USGS	United States geological survey
WAG	water-alternating gas injection
WCI	water curtain injection

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